



SMART SOLUTIONS FOR SMART SERVICES

ISSUE
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PHILIPS

[illegible]

Philips' Smart Label and Tag ICs

Philips' Smart Label and Tag ICs											
Product Features	ICODE 1 SL1 ICS30	ICODE 1 HC* SL1 ICS31	ICODE SLI SL2 ICS20	ICODE EPC SL2 ICS10	ICODE UID SL2 ICS11	UCODE HSL SL3 ICS30	UCODE EPC 1.19	UCODE EPC G2 SL3 ICS10	HITAG™ 1	HITAG™ 2	HITAG™ S
Memory											
Size [bit]	512	512	1024	136	192	2048	96 + 256	512	2048	256	32, 256, 2048
Write Endurance [cycles]	100 000	100 000	100 000	-	10 000	100 000	100 000	100 000	100 000	100 000	100 000
Data Retention [yrs]	10	10	10	5	5	10	10	10	10	10	10
Organisation	16 blocks á 4 bytes	16 blocks á 4 bytes	32 blocks á 4 bytes	17 blocks á 1 byte	24 blocks á 1 byte	64 blocks á 4 bytes	11 blocks á 4 bytes	32 blocks á 2 bytes	64 blocks á 4 bytes	8 blocks á 4 bytes	64 blocks á 4 bytes
RF-Interface											
According to	ICODE 1	ICODE 1	ISO 15693, ISO 18000	EPC	EPC	ISO 18000	ISO 18000	EPC class 1 Gen2	HITAG 1	HITAG 2, ISO 11784/85	HITAG 1+, ISO 11784/85
Frequency	13.56 MHz	13.56 MHz	13.56 MHz	13.56 MHz	13.56 MHz	UHF / 2.45 GHz	UHF / 2.45 GHz	860 - 960 MHz	100 ... 150 kHz	100 ... 150 kHz	100 ... 150 kHz
Baudrate [kbit/s]	up to 26.5	up to 26.5	up to 53	up to 53	up to 53	up to 40	up to 40	up to 640	up to 4	up to 4	up to 8
Anticollision	time slot	time slot	acc. ISO 15693	acc. EPC	acc. EPC	adapted binary tree	adapted binary tree	slotted ALOHA	yes, collision detection	-	yes, collision detection
Operating Distance [m]	up to 1.5	up to 1.5	up to 1.5	up to 1.5	up to 1.5	up to 7.0	up to 7.0	up to 7.0	up to 1.5	up to 1.5	up to 2.0
Security											
Unique Serial Number [byte]	8	8	8	-	5	8	8	4	4	4	4
Write Protection	blockwise	blockwise	blockwise	OTP	-	bytewise	bytewise	blockwise	blockwise	blockwise	blockwise, multi user mode
Access Keys	-	-	-	-	-	-	-	32-bit	32-bit	48-bit	48-bit
Access Conditions	-	-	-	-	-	-	-	Plain, Password	Encrypted Mutual Authentication or Plain	Encrypted Mutual Authentication or Plain	Encrypted Mutual Authentication or Plain
Encryption Algorithm	-	-	-	-	-	-	-	-	yes	yes	yes, for authentication only
Special Features											
EAS	yes	yes	yes	-	-	-	-	yes	-	-	-
AFI	yes	yes	yes	-	-	yes	yes	yes	-	-	-
EPC	-	-	yes	yes	yes	-	yes	yes, 96-bit	-	-	-
TTF Modes	-	-	-	-	-	-	-	-	-	yes	yes
Destroy Command	-	-	-	yes	yes	-	-	yes	-	-	-
Packaging											
Sawn Wafer	-	SL1 ICS3101W/N5D	-	-	-	-	-	-	HT1ICS3002W/N5A	HT2ICS2002W/N5A	-
Sawn Wafer (Au-Bumped)	SL1 ICS3001W/N4D	SL1 ICS3101W/N4D	SL2 ICS2001DW/V4D	SL2 ICS1001DW/V4	SL2 ICS1101DW/V4	SL3 ICS3001W/V4	SL3 ICS3101W/V4	SL3 ICS1001FW/V7A]	-	-	HTSiCHxx01EW/V4**
MOA2 Module	SL1 MOA2S30/D	-	SL2 MOS2001DV	-	-	-	-	-	HT1MOA2S30/E/3	HT2MOA2S20/E/3	HTSMOHxx01EV**
FCP2 Module	-	-	SL2 FCS2001DV/DH	SL2 FCS1001DV/DH	SL2 FCS1101DV/DH	-	-	SL3 FCS1001FV/DH	-	-	HTSFCHxx01EV/DH**
Stick SOT 385-1	-	-	-	-	-	-	-	-	HT1DC20S30/F	HT2DC20S20/F	-
TSSOP8	-	-	-	-	-	SL3S3001FTT	-	SL3 S1001FTT	-	-	-
HVSON2	-	-	-	-	-	-	-	-	-	-	HTSHxx01ETK

* HC: high capacitance (97pF)

** xx is: “32” for 32-bit, “56” for 256-bit, and “48” for 2048-bit memory size

HITAG™ Reader ICs		
Product Features	HTRC110 HITAG™ Reader ICs	PCF7921, HITAG Reader Security Controller
Modulation Type	100 % ASK	not applicable
Dimensions [mm]	6.2 × 8.75 × 1.45	SSOP20
Interface	CMOS	programmable IO Pins
Supply Voltage [V]	5 ±10%	2.1V ... 3.6
Antenna Driver Current [mA]	200 continuous	not applicable
Clock Osc. Frequency [MHz]	4 ... 16	up to 2.2 (single clock instruction)
Operating Temperature [°C]	-40 ... +85	-40 ... +85
Power Down Current [µA typ.]	7	0.1 ; max: 0.5 (RUN: 300 ; IDLE20 ; PD: 100nA)
Memory	-	4 Kbyte EROM (Flash) 512 Byte EEPROM 128 Byte RAM
Supported Products		
HITAG™ 1	yes	-
HITAG™ 2	yes	yes
HITAG™ S	yes	yes
Security		
HITAG™ 1 data encryption	-	-
HITAG™ 2 data encryption	-	yes
HITAG™ S data encryption	-	yes
Package		
SO14, Tube	HTRC110 01T/02EE	-
SO14, Reel	HTRC110 01T/03EE	-
SSOP20	-	PCF7921ATS/3391

Philips' Contact Smart Card Reader ICs

[illegible]

Philips' Contactless Mifare® Smart Card ICs

Product Features	Mifare® Ultralight MF0 IC U1X	Mifare® Std 1k MF1 IC S50	Mifare® Std 4k MF1 IC S70	Mifare® DESFire MF3 IC D40	Mifare® DESFire SAM MF3 IC D40 SAM
Memory					
EEPROM size [byte]	64	1024	4096	4096	72k
OTP area [bit]	32	-	-	-	-
Write Endurance [cycles]	10 000	100 000	100 000	100 000	100 000
Data Retention [yrs]	5	10	10	10	10
Organization	16 pages á 4 byte	16 sectors á 64 byte	32 sectors á 64 byte 8 sectors á 256 byte	flexible file system	128 key entries
RF-Interface					
Acc. to ISO 14443A	yes - up to layer 3	yes - up to layer 3	yes - up to layer 3	yes - up to layer 4	ISO 7616,T=1
Frequency [MHz]	13.56	13.56	13.56	13.56	1 ... 10
Baudrate [kbit/s]	106	106	106	106 ... 424	9.6 ... 1000
Anticollision	bit-wise	bit-wise	bit-wise	bit-wise	-
Operating Distance [mm]	up to 100	up to 100	up to 100	up to 100	-
Security					
Unique Serial Number [byte]	7, cascaded	4	4	7, cascaded	7
Random Number Generator	-	yes	yes	yes, according to FIPS140-2	yes
Access Keys	-	2 keys per sector	2 keys per sector	14 keys per application	128 key entries
Access Conditions	per page	per sector	per sector	per file	per key entry
Mifare® Classic Security	-	supported	supported	-	-
DES & DES3 Security	-	-	-	MACing / Encipherment	MACing / Encipherment
Anti-tear supported by chip	-	for value blocks	for value blocks	yes	-
Special Features					
Multi-application	-	supports MAD*	supports MAD2**	28 applications, MAD3***	-
Purse Functionality	-	Value block format	Value block format	Value file	-
Secure Transport Transaction	48 byte read 16 byte write	512 byte read 16 byte write	512 byte read 128 byte write	512 byte read 128 byte write	-
Transaction Time [ms]	31.4	164	140	105	-
Packaging					
Sawn Wafer	-	MF1ICS5005W/V9D	MF1ICS7001W/V9D	MF3ICD4001DW/V506	-
Sawn Wafer (Au-Bumped)	MF0ICU1X01W/V1D	MF1ICS5005W/V1D	MF1ICS7001W/V1D	MF4ICD4001DW/V406	-
MOA2 Module	-	MF1MOA2S50/D3F	-	-	-
MOA4 Module	-	MF1MOA4S50/D	MF1MOA4S70/D	MF3MOD4001DV/406	-
FCP2 Module	MF0FCP2U1X/DH	MF1FCP2S50/DH	-	-	-
PDM1.1 Module	-	-	-	-	P5CT072EV0/T0M68220

*MAD: Mifare® Application Directory

**MAD2: MAD Extension for 4 kbyte EEPROM size

***MAD3: MAD2 Extension for DESFire

Philips' NFC Devices

Product Features	PN511	PN512	PN531	PN532
Operating distance typ. [mm]	50 (passive mode) 100 (active mode)	50 (passive mode) 100 (active mode)	50 (passive mode) 100 (active mode)	50 (passive mode) 100 (active mode)
Interfaces				
serial [Mbit/s]	up to 1.228	up to 1.228	up to 1.228	up to 1.228
I ² C [bit/s]	400k / 3.4M	400k / 3.4M	400k	400k
SPI [Mbit/s]	up to 5	up to 5	up to 5	up to 5
8-bits parallel	yes	yes	-	-
USB 2.0 (full speed)	-	-	yes	no
CL FIFO depth [byte]	64	64	64	64
Serial/SPI FIFO [byte]	-	-	180	180
S ² C interface	yes	yes	yes	yes
CPU	-	-	80C51	80C51
RAM / ROM [byte]	-	-	1K / 32K	1K / 40K
Integrated LDO voltage regulator	-	-	-	yes
RF Interface				
analog interface	fully integrated	fully integrated	fully integrated	fully integrated
Carrier frequency [MHz]	13.56	13.56	13.56	13.56
Baudrates ISO 18092 [kbits/s]	106 / 212 / 424	106 / 212 / 424	106 / 212 / 424	106 / 212 / 424
Baudrates ISO 14443A [kbits/s]	106 / 212 / 424	106 / 212 / 424	106 / 212 / 424	106 / 212 / 424
Standards & Protocols				
ISO 18092 (active/passive)	yes	yes	yes	yes
ISO 14443A	yes	yes	yes	yes
Mifare Classic	yes	yes	yes	yes
ISO 14443B reader/writer RF interface	-	yes	-	yes
FeliCa RF interface	yes	yes	yes	yes
Security Features				
Mifare classic	yes	yes	yes	yes
Additional Product Information				
Supply voltage [V]	2.5 ... 3.6	2.5 ... 3.6	2.5 ... 4.0	2.5 ... 5.5
Host interface voltage min. [V]	1.6	1.6	1.6	1.6
USB bus power supply [V]	-	-	4.2 ... 5.5	-
Power down mode typ. [μA]	5	5	10	12
Power down mode with	-	-	-	-
RF level detector on [μA]	10	10	30	30
Transmitter supply current typ. [mA]	60	60	60	60
Temperature range [°C]	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85
Packages	HVQFN40/HVQFN32	HVQFN40/HVQFN32	HVQFN40	HVQFN40
Embedded firmware	-	-	yes	yes
Design In Kit	OM5561	OM5571	OM5555	-
	Incl. 2 serial boards plus CD Rom	Incl. 1 serial/ 1 USB board plus CD Rom	Incl. 2 serial boards plus CD Rom	-

Development Tools Overview Philips' Smart Card Controller Families

Product Family	Tool Components					
	Software Tools		Hardware Tools			
	C-Compiler, Assembler, Linker/Locator	Simulator	Source-Level Debugger, Integrated Development Environment	In-Circuit Emulator	Prototyping Kit	Smart Card Probe
Extended 8051 Architecture						
Mifare ProX Family P8RF6xxx P8RF5xxx	Keil Raisonance	Keil * Raisonance *	Ashling Keil Raisonance	Ashling Philips Raisonance	Ashling Raisonance	Ashling Raisonance
SmartMX Family P5Sxyyy P5Cxyyy	Keil	Keil *	Keil Ashling	Ashling Philips	Ashling Keil	Ashling Keil
SmartXA Architecture						
SmartXA2 Family P7Sxyyy P7Cxyyy	Raisonance	Raisonance *	Ashling	Ashling Philips	Ashling	Ashling
Smart MIPS Architecture						
HiPerSmart Family P9Sxyyy	MIPS, Distribution via Ashling	MIPS, Distribution via Ashling	Ashling	Ashling Philips	Ashling	Ashling

* Product specific simulator extensions are available from Philips free of charge

Development Tool Partner Contacts

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0.18µm

Philips' SmartMX Contact and Dual Interface Security Controllers

Product Features	P5SD009	P5SC018	P5SC036	P5SC072
CPU	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51
ISO Contact Interface	ISO 7816	ISO 7816	ISO 7816	ISO 7816
ISO Contactless Interface	ISO 14443	-	-	-
ROM [byte]	96 K	96 K	96 K	128 K
RAM total [byte]	2.25 K	2.25 K	2.25 K	2.25 K
EEPROM [byte]	12 K	18 K	36 K	72 K
Security Features				
DES-Engine	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs
Exception Sensors	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light
Additional Product Information				
UART for implemented interfaces	yes	yes	yes	yes
CRC-Engine	yes	yes	yes	yes
Supply voltage [V]	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5
External Clock [MHz]	1 ... 10 (13.56)	1 ... 10	1 ... 10	1 ... 10
Internal Clock [MHz]	1 ... 30	1 ... 30	1 ... 30	1 ... 30
Temperature range [°C]	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85
EEPROM page mode granularity [byte]	1 ... 64	1 ... 64	1 ... 64	1 ... 64
16-bit Timer/Counter	2	2	2	2
True Random Number Generator	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31
Philips Handshake Technology	yes	yes	yes	yes
Mifare® functionality (optional)	yes, 1K or 4K	-	-	-
Delivery Type Wafer	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn
Delivery Type Module	Contact, Dual, Interface, Contactless	Contact	Contact	Contact
Evaluation and Certification				
3 rd Party Hardware Evaluation	yes	yes	yes	yes
Security Certificates*	CC EAL5+	-	-	-

* Common Criteria (CC), Mastercard, VISA, ZKA evaluations planned depending on application requirements

Philips' SmartMX Contact and Dual Interface PKI Controllers

Product Features	P5CC009	P5CD009	P5CC018	P5CC036	P5CD036	P5CN036	P5CU036	P5CC072	P5CD072	P5CN072	P5CT072
CPU	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51
ISO Contact Interface	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816
ISO Contactless Interface	-	ISO 14443	-	-	ISO 14443	-	-	-	ISO 14443	-	ISO 14443
USB Interface	-	-	-	-	-	-	USB 2.0 (low speed)	-	-	-	USB 2.0 (low speed)
S ² C (NFC) Interface	-	-	-	-	-	yes	-	-	-	yes	-
ROM [byte]	96 K	96 K	128 K	128 K	160 K	160 K	160 K	160 K	160 K	160 K	160 K
RAM (linear addressable) [byte]	4.5 K	4.5 K	4.5 K	4.5 K	4.5 K	4.5 K	4.5 K	4.5 K	4.5 K	4.5 K	4.5 K
- Standard RAM [byte]	3.25 K	3.25 K	3.25 K	3.25 K	3.25 K	3.25 K	3.25 K	3.25 K	3.25 K	3.25 K	3.25 K
- RAM accessible by FameXE [byte]	1.25 K	1.25 K	1.25 K	1.25 K	1.25 K	1.25 K	1.25 K	1.25 K	1.25 K	1.25 K	1.25 K
EEPROM [byte]	10 K	12 K	18 K	36 K	36 K	36 K	36 K	72 K	72 K	72 K	72 K
Security Features											
PKI Crypto-Engine (RSA key length up to 4096-bit)	FameXE	FameXE	FameXE	FameXE	FameXE	FameXE	FameXE	FameXE	FameXE	FameXE	FameXE
RSA Modular Exponentiation M2048 E17-bit (SF)* [ms]	< 35	< 35	< 35	< 35	< 35	< 35	< 35	< 35	< 35	< 35	< 35
ECC (192-bit signature) generation/verification [ms]	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40
DES-Engine	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs
AES-Engine key length 128/192/256-bit [µs]	-	-	-	-	-	11 / 12 / 13	-	11 / 12 / 13	-	11 / 12 / 13	11 / 12 / 13
Exception Sensors	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light
Memory Management Unit (Firewall)	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Additional Product Information											
UART for implemented interfaces	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
CRC-Engine	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Supply voltage [V]	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5
External Clock [MHz]	1 ... 10	1 ... 10 (13.56)	1 ... 10	1 ... 10	1 ... 10 (13.56)	1 ... 6	1 ... 6	1 ... 6	1 ... 10 (13.56)	1 ... 6	1 ... 10 (13.56)
Internal Clock [MHz]	1 ... 30	1 ... 30	1 ... 30	1 ... 30	1 ... 30	1 ... 30	1 ... 30	1 ... 30	1 ... 30	1 ... 30	1 ... 30
Temperature range [°C]	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85
EEPROM page mode granularity [byte]	1 ... 64	1 ... 64	1 ... 64	1 ... 64	1 ... 64	1 ... 64	1 ... 64	1 ... 64	1 ... 64	1 ... 64	1 ... 64
16-bit Timer (also "etu" Counter use)	2	2	2	2	2	2	2	2	2	2	2
True Random Number Generator	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31
Philips Handshake Technology	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Mifare® functionality (optional)	-	yes, 1K or 4K	-	-	yes, 1K or 4K	yes, 1K or 4K	-	-	yes, 1K or 4K	yes, 1K or 4K	yes, 1K or 4K
Delivery Type Wafer	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn
Delivery Type Module	Contact	Contact, Dual Interface, Contactless,	Contact	Contact,	Contact, Dual Interface, Contactless,	-	Contact	Contact,	Contact, Dual Interface, Contactless,	-	Contact, Dual Interface, Contactless,
Delivery Type SMD package	SSOP20	SSOP20	-	SSOP20	-	SSOP20, HVQFN32	-	SSOP20	-	SSOP20, HVQFN32	SSOP20
Evaluation and Certification											
3 rd Party Hardware Evaluation	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Security Certificates**	CC EAL5+ VISA, Cast	CC EAL5+ VISA, Cast	VISA, Cast	CC EAL5+ VISA, Cast	CC EAL5+ VISA, Cast	CC EAL5+	-	CC EAL5+ VISA, Cast	CC EAL5+ VISA, Cast	CC EAL5+	CC EAL5+ VISA, Cast

* Straight Forward

** Common Criteria (CC), Mastercard, VISA, ZKA evaluations planned depending on application requirements

0.18µm

Philips' 32-bit HiPerSmart Smart Computing Platform

	Security Controller
Product Features	P9SC648
CPU	32-bit MIPS RISC CPU
ISO Contact Interface	ISO 7816
ROM [byte]	-
Flash [byte]	512 K
RAM total [byte] *	19 K
Instruction Cache [byte]	2 K
Data Cache [byte]	1 K
EEPROM [byte]	142 K
Security Features	
Exception Sensors	V, f, T, Light
Memory Management Unit	yes
Hardware Firewall	yes
Additional Product Information	
UART for implemented interface	yes
GSM Power Control Mode	yes
Supply voltage [V]	1.62 ... 5.5
External Clock [MHz]	1 ... 6
Temperature Range [°C]	-25 ... +85
EEPROM page mode granularity [byte]	128 bytes
32-bit Timer/Counter	2
True Random Number Generator	yes, according to FIPS140-2 / AIS31
Delivery Type Wafer	sawn/unsawn
Delivery Type Module	-
Evaluation and Certification	
3 rd Party Hardware Evaluation	yes*
Certificates	yes*
Software	
Secure Cryptolibrary	yes*

* Common Criteria (CC), Mastercard, VISA, ZKA evaluations planned
depending on application requirements

0.14µm

Philips' 32-bit HiPerSmart Smart Computing Platform

	Security Controller
Product Features	P9SN288
CPU	32-bit MIPS RISC CPU
ISO Contact Interface	ISO 7816
S2C (NFC) Interface	yes
ROM [byte]	512K
Flash [byte]	-
RAM total [byte] *	19 K
Instruction Cache [byte]	2 K
Data Cache [byte]	1 K
EEPROM [byte]	288 K
Security Features	
DES-Engine	yes
AES-Engine	yes
Exception Sensors	V, f, T, Light
Memory Management Unit	yes
Hardware Firewall	yes
Additional Product Information	
UART for implemented interface	yes
GSM Power Control Mode	yes
Supply voltage [V]	1.62 ... 5.5
External Clock [MHz]	1 ... 6
Temperature Range [°C]	-25 .. +85
EEPROM page mode granularity [byte]	128 bytes
32-bit Timer/Counter	2
True Random Number Generator	yes, according to FIPS140-2 / AIS31
Mifare® functionality (optional)	yes, 1K or 4K
Delivery Type Wafer	sawn/unsawn
Delivery Type Module	-
Evaluation and Certification	
3 rd Party Hardware Evaluation	yes*
Certificates	yes*
Software	
Secure Cryptolibrary	yes*

* Common Criteria (CC), Mastercard, VISA, ZKA evaluations planned
depending on application requirements

Philips' SmartMX Contact and Dual Interface PKI Controllers

Product Features	P5CC020	P5CC040	P5CD040	P5CN080	P5CC080	P5CD080	P5CN144	P5CC144	P5CD144
CPU	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51	Secure_MX51
ISO Contact Interface	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816
ISO Contactless Interface	-	-	ISO 14443	-	-	ISO 14443	-	-	ISO 14443
S ² C (NFC) Interface	-	-	-	yes	-	-	yes	-	-
ROM [byte]	200 K	200 K	200 K	200 K	200 K	200 K	200 K	200 K	200 K
RAM (linear addressable) [byte]	6 K	6 K	6 K	6 K	6 K	6 K	6 K	6 K	6 K
- Standard RAM [byte]	3,5 K	3,5 K	3,5 K	3,5 K	3,5 K	3,5 K	3,5 K	3,5 K	3,5 K
- RAM accessible by FameXE [byte]	2,5 K	2,5 K	2,5 K	2,5 K	2,5 K	2,5 K	2,5 K	2,5 K	2,5 K
EEPROM [byte]	20 K	40 K	40 K	80 K	80 K	80 K	144 K	144 K	144 K
Security Features									
PKI Crypto-Engine (RSA key length up to 8192-bit)	FameXE	FameXE	FameXE	FameXE	FameXE	FameXE	FameXE	FameXE	FameXE
RSA Modular Exponentiation M2048 E17-bit (SF)* [ms]	< 35	< 35	< 35	< 35	< 35	< 35	< 35	< 35	< 35
ECC (192-bit signature) generation/verification [ms]	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40	< 20 / 40
DES-Engine	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs
AES-Engine key length 128/192/256-bit [µs]	11 / 12 / 13	11 / 12 / 13	11 / 12 / 13	11 / 12 / 13	11 / 12 / 13	11 / 12 / 13	11 / 12 / 13	11 / 12 / 13	11 / 12 / 13
Exception Sensors	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light
Memory Management Unit (Firewall)	yes	yes	yes	yes	yes	yes	yes	yes	yes
Additional Product Information									
UART for implemented interfaces	yes	yes	yes	yes	yes	yes	yes	yes	yes
CRC-Engine	yes	yes	yes	yes	yes	yes	yes	yes	yes
Supply voltage [V]	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5
External Clock [MHz]	1 ... 10	1 ... 10	1 ... 10 (13,56)	1 ... 10	1 ... 6	1 ... 10 (13,56)	1 ... 10	1 ... 6	1 ... 10 (13,56)
Internal Clock [MHz]	1 ... 30	1 ... 30	1 ... 30	1 ... 30	1 ... 30	1 ... 30	1 ... 30	1 ... 30	1 ... 30
Temperature range [°C]	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85
EEPROM page mode granularity [byte]	1 ... 64	1 ... 64	1 ... 64	1 ... 64	1 ... 64	1 ... 64	1 ... 64	1 ... 64	1 ... 64
16-bit Timer (also "etu" Counter use)	2	2	2	2	2	2	2	2	2
True Random Number Generator	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31
Philips Handshake Technology	yes	yes	yes	yes	yes	yes	yes	yes	yes
Mifare® functionality (optional)	-	-	yes	yes	-	yes	-	-	yes
Delivery Type Wafer	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn
Delivery Type Module	Contact	Contact	Contact, Dual Interface, Contactless,	-	Contact	Contact, Dual Interface, Contactless,	-	Contact,	Contact, Dual Interface, Contactless,
Delivery Type SMD package	yes	yes	-	-	yes	yes	-	yes	yes
Evaluation and Certification									
3 rd Party Hardware Evaluation	planned	planned	planned	planned	planned	planned	planned	planned	planned
Security Certificates**	CC EAL5+ planned	CC EAL5+ planned	CC EAL5+ planned	CC EAL5+ planned	CC EAL5+ planned	CC EAL5+ planned	CC EAL5+ planned	CC EAL5+ planned	CC EAL5+ planned

* Straight Forward

** Common Criteria (CC), Mastercard, VISA, ZKA evaluations planned depending on application requirements

0.18µm

Philips' 16-bit SmartXA Security and PKI Controllers

	Security Controller		PKI Controller		USB Controller	
Product Features	P7SC073	P7SC145	P7CC073	P7CC145	P7CU073	P7CU145
CPU	16-bit SmartXA2	16-bit SmartXA2	16-bit SmartXA2	16-bit SmartXA2	16-bit SmartXA2	16-bit SmartXA2
ISO Contact Interface	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816	ISO 7816
USB Interface					USB 2.0 (Low Speed)	USB 2.0 (Low Speed)
ROM [byte]	256 K	256 K	256 K	256 K	256 K	256 K
RAM total [byte]	6 K	6 K	7 K	7 K	7 K	7 K
RAM accessible by FameX [byte]	-	-	1056	1056	1056	1056
EEPROM [byte]	72 K	144 K	72 K	144 K	72 K	144 K
Security Features						
Crypto-Engine 32-Bit	-	-	FameX	FameX	FameX	FameX
RSA 1024-bit (SF)*	-	-	270ms	270ms	270ms	270ms
DES-Engine	DES3 < 110 µs	DES3 < 110 µs	DES3 < 110 µs	DES3 < 110 µs	DES3 < 110 µs	DES3 < 110 µs
Exception Sensors	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light
Memory Management Unit	yes, enhanced	yes, enhanced	yes, enhanced	yes, enhanced	yes, enhanced	yes, enhanced
Hardware Firewall	yes	yes	yes	yes	yes	yes
Additional Product Information						
UART for implemented interfaces	yes	yes	yes	yes	yes	yes
UART DMA	yes	yes	yes	yes	yes	yes
CRC-Engine	yes	yes	yes	yes	yes	yes
JAVA VM Support	SmartXAccelerator	SmartXAccelerator	SmartXAccelerator	SmartXAccelerator	SmartXAccelerator	SmartXAccelerator
GSM Power Control Mode	yes	yes	yes	yes	yes	yes
Supply voltage [V]	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5
External Clock [MHz]	1 ... 6	1 ... 6	1 ... 6	1 ... 6	1 ... 6	1 ... 6
Temperature Range [°C]	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85
EEPROM page mode granularity [byte]	1 ... 128	1 ... 128	1 ... 128	1 ... 128	1 ... 128	1 ... 128
16-bit Timer	2	2	2	2	2	2
True Random Number Generator	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31
Delivery Type Wafer	sawn/unsawn	sawn/unsawn	ssawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn
Delivery Type Module	contact	contact	contact	contact	contact	contact
Evaluation and Certification						
3 rd Party Hardware Evaluation	yes	yes	yes	yes	yes	yes
Security Certificates **	VISA	VISA	VISA	VISA	VISA	VISA
Software						
Secure Cryptolibrary	yes	yes	yes	yes	yes	yes

* Straight Forward

** Common Criteria (CC), Mastercard, VISA, ZKA evaluations planned depending on application requirements

Philips' 16-bit SmartXA Security and PKI Controllers

	Security Controller		NFC enabled PKI Controller	
Product Features	P7SC074	P7SC146	P7SN146	P7SN288
CPU	16-bit SmartXA2	16-bit SmartXA2	16-bit SmartXA2	16-bit SmartXA2
ISO Contact Interface	ISO 7816	ISO 7816	ISO 7816	ISO 7816
S ² C Interface	-	-	yes	yes
ROM [byte]	384 K	384 K	384 K	384 K
RAM total [byte]	8 K	8 K	9 K	9 K
RAM accessible by FameX [byte]	-	-	1056	1056
EEPROM [byte]	72 K	144 K	144 K	288 K
Security Features				
PKI Crypto-Engine (RSA key length up to 2048)	-	-	FameXE	FameXE
RSA 1024-bit (SF)*	-	-	270ms	270ms
DES-Engine	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs	DES3 < 50 µs
AES-Engine	-	-	yes	yes
Exception Sensors	V, f, T, Light	V, f, T, Light	V, f, T, Light	V, f, T, Light
Memory Management Unit	yes, enhanced	yes, enhanced	yes, enhanced	yes, enhanced
Hardware Firewall	yes	yes	yes	yes
Additional Product Information				
UART for implemented interfaces	yes	yes	yes	yes
UART DMA	yes	yes	yes	yes
CRC-Engine	yes	yes	yes	yes
JAVA VM Support	SmartXAccelerator	SmartXAccelerator	SmartXAccelerator	SmartXAccelerator
GSM Power Control Mode	yes	yes	yes	yes
Supply voltage [V]	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5	1.62 ... 5.5
External Clock [MHz]	1 ... 6	1 ... 6	1 ... 6	1 ... 6
Temperature Range [°C]	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85
EEPROM page mode granularity [byte]	1 ... 128	1 ... 128	1 ... 128	1 ... 128
16-bit Timer/Counter	2	2	2	2
True Random Number Generator	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31	yes, according to FIPS140-2 / AIS31
Mifare® functionality (optional)	-	-	yes, 1K or 4K	yes, 1K or 4K
Delivery Type Wafer	sawn/unsawn	sawn/unsawn	sawn/unsawn	sawn/unsawn
Delivery Type Module	contact	contact	contact	contact
Evaluation and Certification				
3 rd Party Hardware Evaluation	yes	yes	yes	yes
Security Certificates	-	-	CC EAL 5+**	CC EAL 5 +**
Software				
Secure Cryptolibrary	-	-	yes**	yes**

* Straight Forward

** Common Criteria (CC), Mastercard, VISA, ZKA evaluations planned depending on application requirements



identification
product range

PHILIPS